

2026.03.26.목

🕒 생성일	@2026년 3월 26일 오전 8:54
🏷 태그	Network

▼ 네트워크

- 정적 라우팅 vs 동적 라우팅
- 동적 라우팅 프로토콜 분류

분류	프로토콜	특징
Distance Vector	RIP	홉 수 기준, 소규모
Distance Vector	EIGRP	시스코 전용, 빠름
Link State	OSPF	대규모, 가장 많이 사용
Link State	IS-IS	ISP 환경
Path Vector	BGP	인터넷 전체 라우팅, AS 간 통신

- 네트워크 통신 방식

방식	설명	예시
Unicast	1:1 전송 (특정 1개 대상)	일반 웹 요청, ping
Broadcast	1:전체 전송 (같은 네트워크 전체)	ARP 요청
Multicast	1:그룹 전송 (구독자만)	IPTV, 스트리밍
Anycast	1:가장 가까운 1개 전송	DNS 루트 서버, CDN

▼ 패킷 트레이서 실습

```

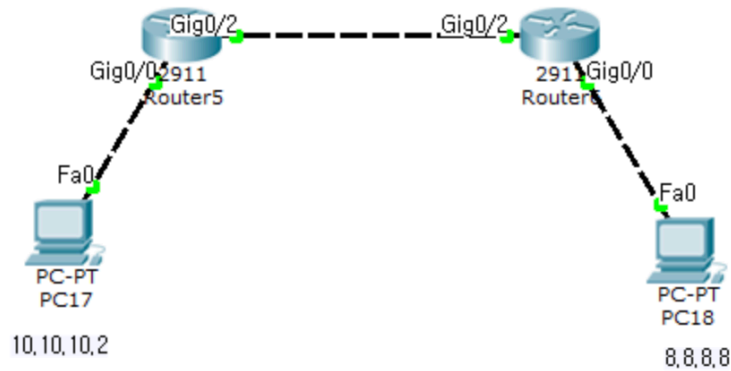
ROUTER1
enable
configure terminal
interface g0/0
ip address 10,10,10,1 255,255,255,0
no shutdown
exit
interface g0/2
ip address 192,168,1,1 255,255,255,0
no shutdown
exit
ip route 8,8,8,0 255,255,255,0 192,168,1,2

```

```

ROUTER2
enable
configure terminal
interface g0/0
ip address 8,8,8,1 255,255,255,0
no shutdown
exit
interface g0/2
ip address 192,168,1,2 255,255,255,0
no shutdown
exit
ip route 10,10,10,0 255,255,255,0 192,168,1,1
end

```



```

PC1
IP: 10,10,10,2
Subnet mask: 255,255,255,0
Gateway: 10,10,10,1

```

```

PC2
IP: 8,8,8,8
Subnet mask: 255,255,255,0
Gateway: 8,8,8,1

```

라우터 3 개를 이용한
PC to PC 연결

ROUTER1
enable
configure terminal

interface g0/0
ip address 10,0,0,1 255,0,0,0
no shutdown

interface g0/1
ip address 1,1,1,2 255,0,0,0
no shutdown

ip route 200,200,200,0 255,255,255,128 1,1,1,1

만약 모두 통신원할시

ip route 200,200,200,0 255,255,255,0 1,1,1,1

만약 ping 2,2,2,2 로 보내고 싶을때

ip route 2,2,2,2 255,255,255,0 1,1,1,1

ROUTER2
enable
configure terminal

interface g0/1
ip address 1,1,1,1 255,0,0,0
no shutdown

interface g0/0
ip address 2,2,2,1 255,0,0,0
no shutdown

ip route 10,0,0,0 255,0,0,0 1,1,1,2

ip route 200,200,200,0 255,255,255,128 2,2,2,8

만약 모두 통신원할시

ip route 200,200,200,0 255,255,255,0 2,2,2,8

ROUTER3

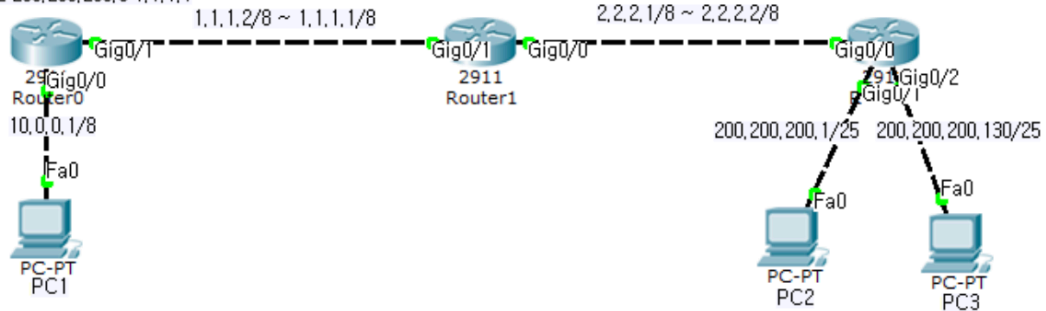
enable
configure terminal

interface g0/0
ip address 2,2,2,2 255,0,0,0
no shutdown

interface g0/1
ip address 200,200,200,1 255,255,255,128
no shutdown

interface g0/2
ip address 200,200,200,254 255,255,255,128
no shutdown

ip route 10,0,0,0 255,255,255,0 2,2,2,1



PC1
IP: 10,10,10,10
Subnet Mask: 255,0,0,0
Gateway: 10,0,0,1

서브넷 정보

200,200,200,0/25 200,200,200,1 ~ 200,200,200,126

PC1 이 속할수있음

200,200,200,128/25 200,200,200,129~200,200,200,254
PC2(130) 와 G0/2 속함

PC1 <-> PC3 연결안됨

PC1 <-> PC2 연결됨

PC2 <-> PC3 연결됨

PC2
IP: 200,200,200,2
Subnet Mask: 255,255,255,128
Gateway: 200,200,200,1

PC3
IP: 200,200,200,130
Subnet Mask: 255,255,255,128
Gateway: 200,200,200,254

라우터, 스위치 이용한
PC to PC 또는 서버 연결

ROUTER 1
enable
configure terminal

interface g0/0
ip address 2,0,0,1 255,0,0,0
no shutdown

interface g0/1
ip address 1,0,0,1 255,0,0,0
no shutdown

ip route 0,0,0,0 0,0,0,0 2,0,0,2
ip route 5,0,0,0 255,0,0,0 2,0,0,2

ROUTER2
enable
configure terminal

interface g0/0
ip address 2,0,0,2 255,0,0,0
no shutdown

interface g0/1
ip address 4,0,0,1 255,0,0,0
no shutdown

interface g0/2
ip address 3,0,0,1 255,0,0,0
no shutdown

ip route 1,0,0,0 255,0,0,0 2,0,0,1
ip route 5,0,0,0 255,0,0,0 4,0,0,2

ROUTER3
enable
configure terminal

interface g0/1
ip address 4,0,0,2 255,0,0,0
no shutdown

interface g0/0
ip address 5,0,0,1 255,0,0,0
no shutdown

ip route 0,0,0,0 0,0,0,0 4,0,0,1
ip route 1,0,0,0 255,0,0,0 4,0,0,1
ip route 3,0,0,0 255,0,0,0 4,0,0,1

